ISE 3200 - HW2: Standard Form & Simplex

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Q1. Convert to standard form: min z = 2x1 - 3x2 subject to x1 + x2 <= 8, 2x1 - x2 >= 2, x1 >= 0, x2 free.

Q1a) Rewrite both constraints as equalities.

x1 + x2 + s1 = 8; 2x1 - x2 - s2 = 2; s1, s2 >= 0

Q1b) Handle the free variable and write the objective in max form.

x2 = x2+ - x2- with x2+, x2- >= 0; max -z = -2x1 + 3x2+ - 3x2-

Q2. Solve with the simplex method: max z = 5x1 + 4x2 subject to 6x1 + 4x2 <= 24, x1 + 2x2 <= 6, x1, x2 >= 0.

Q2a) Set up the initial tableau and choose the entering and leaving variables.

Entering x1 (most negative z-row coefficient, -5); ratios 24/6 = 4 and 6/1 = 6, so s1 leaves.

Q2b) Perform pivots until the tableau is optimal. Show each tableau.

Pivot 1 on row s1; pivot 2: x2 enters, s2 leaves (ratio 2 / (4/3) = 1.5). Final: x1 = 3, x2 = 1.5.

Q2c) State the optimal solution, the optimal value and which constraints are binding.

x1 = 3, x2 = 1.5, z = 21; both constraints binding (s1 = s2 = 0).

Q3) A refinery blends light (L) and heavy (H) crude into 1,000 barrels of fuel. At least 40% must be light crude, light costs $60/bbl and heavy $45/bbl, and at most 700 bbl of light crude is available. Formulate a minimum-cost LP.

min 60L + 45H s.t. L + H = 1000; L >= 0.4(L + H); L <= 700